MODEL comparision matrix

Classfication	Accuracy	Precision	Recall	F1Score
Logistic Regression	0.8554	0.7364	0.6224	0.6747
Decision Tree Classifier	0.8127	0.6074	0.6276	0.6173
Random Forest Classifier	0.8567	0.7347	0.6335	0.6804
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.8692	0.7922	0.6190	0.6950
K-Nearest Neighbors Classifier	0.8243	0.6506	0.5842	0.6156
Support Vector Classifier (SVC)	0.8583	0.7636	0.5961	0.6695
Voting vs Average (best 3 out of 6)	NA	NA	NA	NA
Ensemble	NA	NA	NA	NA

Which model gave you the best result and which was the worst?

Based on the results in the comparison table, Gradient Boosting performed the best overall, achieving the highes score and ROC-AUC on both the validation and test sets. This indicates strong predictive performance and good generalization.

The Voting Ensemble also performed well, closely matching Gradient Boosting. By combining multiple models, it stable and consistent predictions, although it did not outperform the best individual model.

Random Forest was another strong performer, with high ROC-AUC and balanced F1-score, making it a reliable alt

Logistic Regression performed well overall, though it was slightly less effective than tree-based models in capturi complex relationships in the data.

KNN performed moderately, while SVC showed improved performance after scaling but still did not outperform t models. Naive Bayes performed the worst overall, with significantly lower F1-score and recall due to its assumpti feature independence.

Based on overall performance, Gradient Boosting is the recommended model for this task. It provides the best ba accuracy, F1-score, and ROC-AUC. The Voting Ensemble is a strong alternative, but did not significantly improve performance over Gradient Boosting.

Regression	MAE	MSE	Rsquared
Linear Regression			
Decision Tree Regressor			
Random Forest Regressor			
Gradient Boosting Regressor (e.g., XGBoost or LightGBM)			
K-Nearest Neighbors Regressor			
Voting Vs Avearge (best 3 out of 6)			
Ensemble			

Which model gave you the best result and which was the worst?

Naive Bayes performed the worst overall, with significantly lower F1-score and recall due to its assumption of

ROC/AUC
0.9094
0.7495
0.9077
0.9248
0.8470
0.9006
NA
NA

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